CS1231S: Discrete Structures

Tutorial #3: Sets

(Week 5: 12 – 16 Feb 2024)

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1. Discussion Questions

These are meant for your own exercise. No answers will be provided.

D1 Which of the following are true? Which are false?

(a) $\emptyset \in \emptyset$.

(b) $\emptyset \subseteq \emptyset$.

(c) $\emptyset \in \{\emptyset\}$.

(d) $\emptyset \subseteq \{\emptyset\}$.

(e) $\{\emptyset, 1\} = \{1\}$.

(f) $1 \in \{\{1,2\}, \{2,3\}, 4\}$.

(g) $\{1,2\} \subseteq \{3,2,1\}$.

(h) $\{3,3,2\} \subsetneq \{3,2,1\}$.

(i) $\emptyset \in \{1,2\}$.

(j) $\{1,2,3\} = \{3,1,1,2\}$.

(k) $\{5\} \in \{2,5\}$.

(l) $\{1,2\} \in \{1, \{2,1\}\}$.

D2. Let $A = \{1, \{1,2\}, 2, \{2,1,1\}\}$. What is $|A|$?

D3. Let $A = \{0,1,4,5,6,9\}$ and $B = \{0,2,4,6,8\}$. Find $|A \cap B|$ and $|A \cup B|$.

2. Tutorial Questions

Note that the sets here are finite sets, unless otherwise stated.

Have you done the discussion questions above? Make sure you have all the answers correct! (How do you check? Discuss with your friends or post your answers on Canvas or QnA.)

1. Let $\mathcal{P}(A)$ denotes the power set of A. Find the following:

   (a) $\mathcal{P}(\{a, b, c\})$;

   (b) $\mathcal{P}\left(\mathcal{P}\big(\mathcal{P}(\emptyset)\big)\right)$.

2. Let $A = \{5,6,7, \dots, 12\}$. Find the following:

   (a) $\{n \in A : n \text{ is even}\}$;

   (b) $\{n \in A : n = m^2 \text{ for some } m \in \mathbb{Z}\}$;

   (c) $\{-5, -4, -3, \dots, 5\} \setminus \{1,2,3, \dots, 10\}$;

   (d) $\overline{\{5,7,9\} \cup \{9,11\}}$, where A is considered the universal set;

   (e) $\{(x, y) \in \{1,3,5\} \times \{2,4\} : x + y \geq 6\}$.

3. (AY2022/23 Sem2 mid-term test.)
For each of the following statements, prove whether it is true or false.

(a) There exist non-empty finite sets A and B such that $|A \cup B| = |A| + |B|$.

(b) There exist non-empty finite sets A and B such that $|A \cup B| \neq |A| + |B|$.

(c) There exist finite sets A and B such that $A \times \mathcal{P}(B) = \mathcal{P}(A \times B)$.

Aiken thought that (c) is false and he provided a proof as follows:

1. Let $|A| = n, |B| = k$.
2. Then $|\mathcal{P}(B)| = 2^k$ (cardinality of power set).
3. $|A \times B| = nk$ (cardinality of Cartesian product) and hence $|\mathcal{P}(A \times B)| = 2^{nk}$ (cardinality of power set).
4. Then $|A \times \mathcal{P}(B)| = n2^k$ (cardinality of Cartesian product).
5. Since $|A \times \mathcal{P}(B)| = n2^k \neq 2^{nk} = |\mathcal{P}(A \times B)|$, therefore the statement is false.

Is Aiken's conclusion that "the statement is false" correct? Is his proof correct? If his proof is not correct, can you write a correct proof?

4. Let $A = \{2n + 1 : n \in \mathbb{Z}\}$ and $B = \{2n - 5 : n \in \mathbb{Z}\}$. Is $A = B$? Prove that your answer is correct. What does this tell us about how odd numbers may be defined?

5. Using definitions of set operations (also called the **element method**), prove that for all sets A, B, C,

$$A \cap (B \setminus C) = (A \cap B) \setminus C.$$

6. (Past year's midterm test question.)
Using **set identities** (Theorem 6.2.2), prove that for all sets A, B and C,

$$A \setminus (B \setminus C) = (A \setminus B) \cup (A \cap C).$$

7. For sets A and B, define $A \oplus B = (A \setminus B) \cup (B \setminus A)$.

(a) Let $A = \{1,4,9,16\}$ and $B = \{2,4,6,8,10,12,14,16\}$. Find $A \oplus B$.

(b) Using set identities (Theorem 6.2.2), prove that for all sets A and B,

$$A \oplus B = (A \cup B) \setminus (A \cap B).$$

8. Let A and B be set. Show that $A \subseteq B$ if and only if $A \cup B = B$.

9. Hogwarts School of Witchcraft and Wizardry where Harry Potter attended was divided into 4 houses: Gryffindor, Hufflepuff, Ravenclaw and Slytherin.

Let $HSWW$ be the set of students in the Hogwarts School of Witchcraft and Wizardry, and G, H, R and S be the sets of students in the 4 houses.

What are the necessary conditions for $\{G, H, R, S\}$ to be a partition of $HSWW$? Explain in English and the write logical statements.

For questions 10 and 11, for sets $A_m, A_{m+1}, \dots, A_n$, we define the following:

$$\bigcup_{i=m}^{n} A_i = A_m \cup A_{m+1} \cup \dots \cup A_n$$

and

$$\bigcap_{i=m}^{n} A_i = A_m \cap A_{m+1} \cap \dots \cap A_n$$

10. Let $A_i = \{x \in \mathbb{Z} : \ x \geq i\}$ for all integers i. Write down $\bigcup_{i=2}^{5} A_i$ and $\bigcap_{i=2}^{5} A_i$ in **roster notation**.

11. Let $V_i = \left\{x \in \mathbb{R} : \ -\frac{1}{i} \leq x \leq \frac{1}{i}\right\} = \left[-\frac{1}{i}, \frac{1}{i}\right]$ for all positive integers i.
    (a) What is $\bigcup_{i=1}^{4} V_i$?
    (b) What is $\bigcap_{i=1}^{4} V_i$?
    (c) What is $\bigcup_{i=1}^{n} V_i$, where n is a positive integer?
    (d) What is $\bigcap_{i=1}^{n} V_i$, where n is a positive integer?
    (e) Are $V_1, V_2, V_3, \dots$ mutually disjoint?

12. Consider the claim:

    For all sets A, B, C, $\quad (A \setminus B) \cup (B \setminus A) = (A \cup B) \setminus (A \cap B)$.

    Aiken wrote the following proof:

    1. $x \in (A \setminus B) \cup (B \setminus A)$
    2. $\Rightarrow x \in (A \setminus B)$ or $x \in (B \setminus A)$
    3. $\Rightarrow x \in A$ and $x \notin B$ or $x \in B$ and $x \notin A$
    4. $\Rightarrow x \in A$ or $x \in B$ and $x \notin A$ and $x \notin B$
    5. $\Rightarrow x \in (A \cup B)$ and $x \in (\overline{A \cap B})$
    6. $\Rightarrow x \in (A \cup B) \cap (\overline{A \cap B})$
    7. $\Rightarrow x \in (A \cup B) \setminus (A \cap B)$.

    Therefore, $(A \setminus B) \cup (B \setminus A) = (A \cup B) \setminus (A \cap B)$.

    (a) What is wrong with Aiken's proof?
    (b) Prove or disprove the claim.